Patentability Report: Collapsible Bicycle Helmet with Twist-Lock Honeycomb Liner

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Researcher: Hermes Agent
Status: Directional read not a patentability opinion
Version: 3 (final comprehensive)

Invention Summary

A collapsible bicycle helmet comprising:

1. **Honeycomb protective liner** made of overlapping thermoplastic cells
2. **Outer shell + inner honeycomb liner** (two-part construction)
3. **Quarter-turn twist actuation**: shell rotates ~90 relative to liner cells lock into rigid expanded configuration
4. **Reverse twist**: cells collapse flat for compact storage
5. **Radial cell arrangement** around the head
6. **Impact absorption** through progressive cell wall crumpling

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Element-by-Element Prior Art Matrix

#	Element	Status	Closest Reference	Disclosure
1	Honeycomb liner	**Anticipated**	WO2023233224A1, US12268268B2, CN122439960A, CN207285346U, EP4179908A1	Honeycomb protective configuration, radial cell direction, staggered layers
2	Overlapping/staggered cells	**Anticipated**	WO2023233224A1	"layers joined to adjacent layers along **staggered** lines of attachment"
3	Locks rigid when actuated	**Anticipated**	WO2023233224A1, US9743704B2, US11857020	End plates engage to form rigid loop; bosses/catches lock; locking clip
4	Collapses flat	**Anticipated**	WO2023233224A1, US12268268B2	"collapsed (flattened) form"; "collapsed or folded condition"
5	Impact absorption by cell crush	**Anticipated**	WO2023233224A1 + 15+ papers	"absorbing impact energy through progressive crumpling"
6	**Quarter-turn twist actuation**	**Not found in helmets**	US11857020 (180 rotation, not 90)	Crown rotates 180 relative to base; different mechanism
7	**Quarter-turn twist mechanism (general)**	**Well-known in mechanical engineering**	US4893978, US6575682, US9200660, US12736079, US20110110716A1, US4653970, US5123795	Quarter-turn fasteners, bayonet locks, twist-lock couplings ubiquitous

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Key Patent References (Ranked by Relevance)

Helmet-Specific Prior Art

Rank	Patent	Title	Overlap
1	**WO2023233224A1** / US12685363	Collapsible helmet, honeycomb, end plates engage to form rigid loop	**Very high** all elements except twist actuation
2	**US12268268B2** / US12740611	Machine vendible expandable helmet, radial honeycomb matrix	**Very high** honeycomb + fold/expand + side frames
3	**US11857020**	Collapsible protective helmet, crown rotates 180 relative to base	**High** rotation between helmet components + locking
4	**US11191316** / US11589634	Collapsible helmet, center + side chassis rotate to collapse	**High** rotation to collapse + locking mechanism
5	**CN122439960A**	Foldable helmet, honeycomb + airbag auto-unfold	**High** honeycomb + auto-unfold
6	**US9743704B2** / US9867416B2**	Collapsible helmet, hinged panels, bosses/catches lock	**Medium** locking mechanism concept
7	**CN207285346U**	Bicycle helmet with flexible honeycomb matrix	**Medium** base honeycomb helmet
8	**EP4179908A1**	Cellular liners in foam shell	**Medium** cellular liner retention
9	**WO2018200414A1** (Fend)	Collapsible helmet	**Medium** commercial folding helmet
10	**US10905187B1** (Gwenventions)	Collapsible helmet	**Medium** commercial folding helmet
11	**GB2567411A**	Folding helmet interlocking forms on mating surfaces	**Medium** interlocking mating surfaces

| 12 | **EP3558045NWB1** | Helmet comprising a segmented shell (CPC A42B 3/32) | **Medium** segmented shell |

Helmet Rotation/Locking Mechanisms

Patent	Title	Relevance
CA2561540C	Adjustment mechanism for helmet sliding shell sections with pivot lock	Shell sections slide + pivot lock (A42B 3/32)
CA2970532A1	Protective helmet systems enabling helmet to rotate independent of head	Shell rotates relative to head/liner
EP3391765A3	Safety helmet with rotary impact buffering shell rotates relative to liner	Shell-liner rotation for impact buffering
WO2003026452A1	Locking device for adjustable helmets cam/lever locking	Adjustable helmet with locking device
US20070056071A1	Firefighter's helmet inner/outer shell retaining mechanism	Inner/outer shell with retaining mechanism
CA3096994A1	Adjustable helmet assembly front/back shell with cam locking	Cam locking mechanism for helmet shells
WO2023196826A1	Assembly for releasably connecting inner helmet and outer helmet shell	Inner/outer helmet connection system

Quarter-Turn Twist-Lock Mechanisms (General Mechanical Engineering)

Patent	Title	Relevance
US4893978	Quarter-turn detent twistlock	Square bore + latching shoulders, 90 rotation
US6575682	Quarter turn fastener	90 rotation, cam-stop, blade member
US9200660	Quarter turn fastener	Lock arms + lock ribs, 90 rotation, stop ribs
US12736079	Quarter turn retainer	90 rotation, edge profile for quarter-turn
US20110110716A1	Quarter turn locking mechanism	Bayonet catch, 90 rotation, spinal surgery tool
US4653970	Quarter-turn fastener	Spiral cam slot, 1/4 turn locking
US5123795	Quarter turn fastener	Slot + tab, 90 rotation, spring-loaded
US6581986B2	Bayonet locking system for vending machines	90 rotation + axial pull
US9353897	Bayonet fastening for pipeline components	30-50 rotation, latching lug

Design Patents (Folding/Collapsible Helmets)

Patent	Title	Assignee
USD774256S1	Folding helmet	Chris J. Katopis
USD873495S1	Collapsible helmet	Fend Corp.
USD759899S1	Bicycle helmet	
USD903947S1	Bicycle helmet	
USD903946S1	Bicycle helmet	
USD1042978S1	Bicycle helmet	

Full Claims Analysis: Top 3 Patents

1. WO2023233224A1 (Monopro Ltd) The Killer Reference

Key claim language:
> "A collapsible helmet deployable for protecting a head of a user, the helmet comprising: an arcuate layered structure openable into a honeycomb protective configuration for encompassing at least an upper part of the head, the arcuate layered structure being formed from layers joined to adjacent layers along staggered lines of attachment so as to define, when opened, a honeycomb structure, wherein the staggered lines of attachment are non-straight lines."

Critical features:
- Honeycomb protective configuration
- Staggered lines of attachment
- Arcuate end plates engage to form rigid loop
- Locking angle > 180 (preferably 220)
- Collapsed (flattened) form
- Impact absorption through progressive crumpling

What it does NOT disclose:
- Quarter-turn twist actuation (uses manual stretching to engage/disengage end plates)
- Shell rotating relative to liner

2. US12268268B2 (Machine Vendible Expandable Helmet)

Key claim language:
> "A bicycle helmet that fits over a surface of a head of a user, the bicycle helmet comprising: at least one segment of flexible cell structures that form a radial honeycomb matrix movable between a folded condition where each side of the at least one segment is disposed generally parallel and an expanded condition where the radial honeycomb matrix of the at least one segment is configured to be expanded at least partially over the head of the user and arranged radially relative to the surface of the head of the user."

Critical features:
- Radial honeycomb matrix
- Folded expanded
- Side frames at ends
- Sliding latch for head size adjustment
- "first and second frames rotate to a position beyond parallel in the expanded use position"

What it does NOT disclose:
- Quarter-turn twist actuation (uses chin strap pull to expand)
- Shell rotating relative to liner

3. US11857020 (Coco Worldwide Ltd) The Rotation Reference

Key claim language:
> "A collapsible protective helmet comprising: an impact-resistant base body configured to encircle the head of a wearer, an impact-resistant crown defining a rim portion, coupling means for coupling the crown to the base body to be movable between a protective position and a stored position, and releasable locking means for locking the base body and crown together in each of the two positions of the crown."

Critical features:
- Crown rotates 180 relative to base body
- Ball-and-socket joint for rotation
- Over-centre toggle latch for locking
- Honeycomb structure in base body and/or crown
- Collapsed volume reduction > 40%

What it does NOT disclose:
- Quarter-turn (90) rotation uses 180 rotation
- Shell rotating relative to liner uses crown rotating relative to base body
- Twist-lock mechanism uses pivot + toggle latch

Non-Patent Literature (OpenAlex/Crossref, 42 refs)

Key papers:
- "Optimisation of an elastomeric pre-buckled honeycomb helmet liner for advanced impact mitigation" (2023, cited 15)
- "Viscoelastic circular cell honeycomb helmet liners for reducing head rotation and brain strain in oblique impacts" (2024, cited 14)
- "Evaluation of a novel bicycle helmet concept in oblique impact testing" (2019, cited 99)
- "Designing for Sustainability and Safety in Urban Micro-mobility: A Novel Helmet Concept" (2023)
- "Modelling and Analysis of Foldable Motorcycle Helmet for Human Comfort" (2020)
- "A Biomechanical Evaluation of a Novel Airbag Bicycle Helmet Concept" (2021, cited 19)
- "Helmet Design Based on the Optimization of Biocomposite Energy-Absorbing Liners" (2019, cited 92)
- "An Overview of the Effectiveness of Bicycle Helmet Designs in Impact Testing" (2021, cited 25)

RETRACTED (excluded from analysis):
- "Structural improvement of 3D woven honeycomb composite liner" (Research Square, retracted)

The Twist Actuation: Deep Analysis

What I found in helmet patents:
- **US11857020**: Crown rotates **180** relative to base body (not 90), using a pivot + locking clip. Different mechanism the crown flips over, not a twist-lock.
- **US11191316/US11589634**: Side chassis rotate relative to center chassis (not shell vs. liner).
- **WO2023233224A1**: End plates engage by manually stretching the structure (not twisting).
- **US9743704B2**: Panels hinge and snap-lock with bosses/catches (not twist).
- **CA2561540C**: Shell sections **slide** (not rotate) with pivot lock.

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- **EP3391765A3**: Shell rotates relative to liner but for **impact buffering**, not actuation.
- **CA2970532A1**: Shell rotates relative to head for **impact decoupling**, not actuation.
- **GB2567411A**: Folding helmet with interlocking mating surfaces uses hinge + fixings, not twist.

What I found in general mechanical engineering:
- Quarter-turn twist-lock mechanisms are **extremely well-known**:
- US4893978: "one-quarter turn twistlock fastener with very positive quarter turn latching positions"
- US6575682: "quarter turn fastener... rotated through an arcuate extent of up to approximately 90"
- US9200660: "quarter turn fastener... rotated 90 degrees clockwise"
- US12736079: "quarter turn retainer... rotated approximately 90 degrees"
- US20110110716A1: "quarter turn locking mechanisms... bayonet catch"
- US4653970: "quarter-turn fastener... spiral cam slot... 1/4 turn"
- US5123795: "quarter turn fastener... one fourth of the way around the circumference"
- Plus: bayonet couplings, camera lens mounts, twist-lock bottle caps, etc.

What I did NOT find:
- A patent disclosing a **quarter-turn (90) twist** of the shell relative to the liner to lock/unlock a honeycomb structure **in a helmet**.

But:
- Quarter-turn twist-lock mechanisms are **ubiquitous in mechanical engineering** (dozens of patents, 1970s-2020s)
- Applying a known mechanical principle to a known structure is **obvious** under 35 U.S.C. 103 (KSR v. Teleflex)
- The US11857020 patent already teaches rotation between helmet components as an actuation mechanism
- An examiner would likely combine US11857020 (helmet rotation) + US4893978/US6575682 (quarter-turn lock) reject under 103

Design Patent Landscape

Six design patents found for folding/collapsible helmets:
- **USD774256S1** (Katopis, 2016) folding helmet
- **USD873495S1** (Fend Corp, 2020) collapsible helmet
- **USD759899S1**, **USD903947S1**, **USD903946S1**, **USD1042978S1** bicycle helmets

Design patents protect **ornamental appearance**, not function. If your helmet has a unique visual appearance (e.g., visible twist-lock mechanism, unique honeycomb pattern, distinctive folded shape), a design patent may be viable alongside or instead of a utility patent.

Honest Verdict

Core concept is heavily anticipated. The combination of:
- Honeycomb liner
- Overlapping/staggered cells
- Locks rigid when actuated
- Collapses flat
- Impact absorption by cell crush

...is disclosed in **WO2023233224A1** and the **US12268268B2/US12740611** family. This is a crowded, commercially active field (Closca, FEND, Overade PLIXI, Newlane all sell folding helmets).

What survives as potentially novel: the specific **quarter-turn twist actuation** shell rotating 90 relative to liner to switch rigidcollapsed. No patent found disclosing this exact mechanism in a helmet. However:

1. It's a **narrow claim** at best
2. It's **likely obvious** in view of:
- US11857020 (rotation between helmet components)
- US4893978 / US6575682 / US9200660 (quarter-turn twist-lock fasteners decades of prior art)
- KSR v. Teleflex (applying known mechanical principle to known structure = obvious)
3. An examiner would likely reject it under 103 as an obvious combination
4. The **specific geometry** of how the twist actuates the honeycomb cells (cam surfaces, detent positions, etc.) might be the only protectable aspect but this would be a very narrow design patent or utility patent claim

This is a directional read, not a patentability opinion. For a real filing decision you need a classification-driven search (CPC **A42B 3/125**, **A42B 3/32**, **A42B 3/128**) by a registered patent attorney.

Search Methodology

- **Sources:** web_search (Hermes backend), web_extract (Google Patents), OpenAlex, Crossref
- **Patent77 search:** Rate-limited (daily quota of 20 searches reached)
- **Patent77 get_patent:** Upstream unavailable (rate-limits heavy use)
- **Direct curl:** IP-throttled by Google (503)
- **Queries:** 4 patent framings + 2 literature framings + 2 twist-lock framings + 1 CPC classification + 1 design patent
- **Total unique refs:** 42 NPL + 12 helmet patents + 7 helmet rotation patents + 9 quarter-turn patents + 6 design patents = **76 total**
- **Search date:** 2026-10-08

Commercial Products in the Space

- **Closca** folding honeycomb commuter helmet (89.90)
- **Overade PLIXI** foldable urban bike helmet
- **FEND One** folding helmet, 50% size reduction
- **Newlane** collapsible cycling helmet with Flip-Clip
- **HIVE** origami-inspired collapsible bike helmet (student project)

Recommended Next Steps

1. **Classification-driven search** by a registered patent attorney (CPC A42B 3/125, A42B 3/32, A42B 3/128)
2. **Deep search on twist actuation geometry** search for specific cam/detent geometries in helmet contexts
3. **Consider design patent** instead of utility patent the visual appearance of a twist-lock folding helmet may be protectable (6 design patents already exist in this space)
4. **Provisional filing** if you want to preserve a priority date while continuing research
5. **Consider the specific honeycomb geometry** if your cell geometry is novel (e.g., specific staggered pattern, specific cell shape), that might be independently patentable
6. **Patent77 retry tomorrow** their daily quota resets; their AI search may find references my keyword search missed